

Physics Factsheet



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Specific Heat Capacity and Specific Latent Heat

Heat Energy

When an object is either supplied with heat energy, or it loses heat energy, there are two possible consequences for the object, a change in its **temperature** and a change of **state** (from liquid to gas for example). This factsheet is concerned with understanding and describing these two consequences of heat energy transfer.

The symbol used for heat energy is Q .

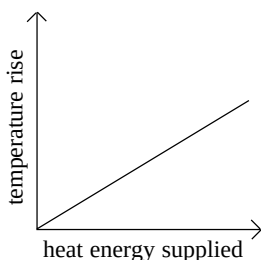
The two consequences of a change in heat energy:

Changes in the heat energy of an object can lead to a change in temperature of the object, a change of state of the object or a combination of both of these things.

- If an object is **heating up** it has had an **increase** in heat energy and it is said to have had a **positive** change in heat energy Q will be a **positive** number.
- If an object is **cooling down** its heat energy is **decreasing** and it is said to have had a **negative** change in heat energy Q will be a **negative** number.

Specific Heat Capacity – changes in temperature.

From everyday experience of heating substances, such as heating up water for a cup of coffee, it is easy to believe that the more heat energy that a substance receives, the larger its temperature rise. In fact accurate experiments would show that if you double the amount of heat energy that you give to a material, you would double the increase in temperature of the substance. A graph of temperature rise against energy supplied would be a straight line through the origin, as shown below.



This shows that temperature rise and heat energy supplied are directly proportional to each other.

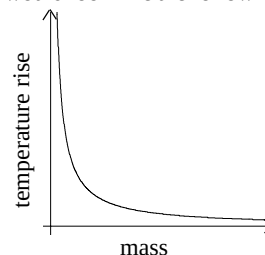
The symbol for temperature rise is $\Delta\theta$. (The θ symbol represents temperature and the Δ symbol represents a change.) An **increase** in temperature is given a **positive** value.

Mathematically we can express that change in temperature is proportional to the heat energy supplied as follows:

$$\Delta\theta \propto Q$$

Similarly, it is easy to understand that if you take greater and greater masses of a substance and supply each mass with the same amount of heat energy, they will have smaller and smaller increases in temperature. This can be seen through every day experience as well; a kettle may boil a cup full of water in a few minutes, but if the kettle is full of water and you still only switch it on for a few minutes, the water may only be lukewarm. More accurate experiments reveal that if you double the mass of the

material being heated, you will only get half the temperature rise if the same amount of heat energy is supplied. A graph of temperature rise against mass of material would look like the following:



This type of relationship shows that temperature rise is inversely proportional to the mass of the material being heated. Mathematically, we can express that temperature rise is inversely proportional to the mass of material being heated as:

$$\Delta\theta \propto \frac{1}{m}$$

The factors affecting the temperature rise of a material.

As the heat energy supplied to a constant mass of material is increased, the temperature rise of the material increases.

- Energy supplied and temperature rise are directly proportional.
- As the mass of a material increases the temperature rise decreases for a constant amount of energy supplied.
- Mass of material and temperature rise are inversely proportional.

The two proportionalities given above can be combined into one:

$$\Delta\theta \propto \frac{Q}{m}$$

This combined proportionality can be turned into an equation by inserting a constant of proportionality, c .

$$c\Delta\theta = \frac{Q}{m}$$

The constant of proportionality c is known as the **specific heat capacity**.

Definition of specific heat capacity

Specific heat capacity is the amount of heat energy needed to increase the temperature of 1 kg of the material by 1 °C.

$$c = \frac{Q}{m\Delta\theta}$$

c = specific heat capacity ($\text{Jkg}^{-1}\text{°C}^{-1}$)

Q = heat energy supplied (J)

m = mass of material being heated (kg)

$\Delta\theta$ = increase in temperature (°C)

Typical exam question

An immersion heater is to heat 50kg of water contained in a copper hot water tank of mass 12 kg.

(a) Calculate how much energy is required to increase the temperature of the water from 20°C to its boiling point. (2)

(b) What assumption have you made in the calculation for part (a)? (1)
Specific heat capacity of water = $4200 \text{ Jkg}^{-1}\text{°C}^{-1}$

(a) Increase in temperature, $\Delta\theta$ = final temperature – initial temperature
= $100 - 20 = 80 \text{ °C}$

Heat energy supplied, $Q = mc\Delta\theta = (50)(4200)(80) \checkmark = 16,800,000 \text{ J} \checkmark$

(b) That none of the water has evaporated ✓

Exam Hint: - Don't be concerned when your answers to calculations seem to be large numbers for the amount of heat energy needed for heating up materials such as water. It requires lots of energy to increase the temperature of some materials.

Specific heat capacity and cooling down

The equation for specific heat capacity is not confined to heating up materials, it can also be used when materials cool down. It is important to remember that when a material is cooling down it is losing heat energy so the value for *heat energy supplied*, Q , will be negative. Similarly, the temperature of the material will be falling, and the value for temperature increase, $\Delta\theta$, will be negative.

Energy supplied when cooling down

When a material is cooling down its total amount of heat energy is decreasing, and any value for heat energy "supplied", Q , will be negative. This will also mean a negative value for increase in temperature, $\Delta\theta$, as the temperature of the material is decreasing.

Worked Example: An oven tray of mass 120g and made of iron is taken out of the oven and cools by losing 8,000 J of heat energy. When the tray is initially at a temperature of 200°C what will be its final temperature? The specific heat capacity of iron = 470 Jkg⁻¹°C⁻¹

A rearranged specific heat capacity equation can be used to find the increase in temperature for the tray but it is important to include the negative sign in front of the value for heat energy supplied. Also note how the mass of the tray has been converted into kilograms before it has been substituted into the equation.

$$\Delta\theta = Q/mc = (-8000)/(0.12)(470) = -142^\circ\text{C}$$

This has now given us a negative value for temperature rise, as we would expect because the tray is cooling.

$$\text{Final temperature} = 200 - 142 = 58^\circ\text{C}$$

Specific latent heat – changes in state

If a material is going through a change of state, from solid to liquid, liquid to gas or vice versa, then its temperature will remain constant. The material will still require heat energy to melt it or boil it, but the heat energy is used to alter the arrangement of the molecules of the material. The spacing between the molecules is increased and work is done against the forces of attraction between the molecules. Similarly, the material will also give out heat energy if it is condensing or freezing.

Our existing equation for specific heat capacity **cannot** be used for changes of state.

The more heat energy that is supplied to a material that is changing state, the more of that material will change. Heat energy supplied, Q , and mass of material changing state, m , are directly proportional.

$$Q \propto m$$

Once again, this proportionality can be changed into an equation by inserting a constant of proportionality, l .

$$Q = lm$$

Q = Heat energy supplied (J)

m = Mass of material changing state (kg)

The constant of proportionality, l , is called the specific latent heat and it has the units Jkg⁻¹.

Every material has two values of specific latent heat, one value for a change of state from a solid to a liquid, called the specific latent heat of fusion ("fusion" is another word for "melting"), and one value for a change of state from a liquid into a gas, called the specific latent heat of vaporisation.

The symbol used for both specific latent heat of fusion and specific latent heat of vaporisation is l .

Definition of specific latent heat

Specific latent heat of fusion is defined as the amount of energy required to turn 1 kg of a solid into a liquid at a constant temperature.

Specific latent heat of vaporisation is defined as the amount of energy required to turn 1 kg of a liquid into a gas at a constant temperature.

$$l = Q/m$$

l = specific latent heat (Jkg⁻¹)

Q = heat energy supplied (J)

m = mass of material changing state (kg)

Gaining and losing heat energy when changing state

- If a substance is changing from a solid to a liquid or from a liquid to a gas then heat energy is being supplied to a substance and Q is positive.
- If a substance is changing from a gas to a liquid or a liquid to a solid then it is losing heat energy and the heat energy "supplied", Q , is negative.

Typical exam question

A kettle is filled with 2.0kg of water.

- Calculate how much energy is required to increase the temperature of the water from 20°C to boiling point, 100°C. (3)
- If the kettle is left running so that it supplies another 12000J of energy, how much of the water will have evaporated and turned from a liquid into a gas. (2)

$$\text{specific heat capacity of water} = 4200 \text{ Jkg}^{-1}\text{°C}^{-1}$$

$$\text{specific latent heat of vaporisation of water} = 2.3 \times 10^6 \text{ Jkg}^{-1}$$

- Change in temperature, $\Delta\theta$ = final temperature – initial temperature
= 100 – 20 = 80 °C ✓
Heat energy supplied, $Q = mc\Delta\theta = (2)(4200)(80)$ ✓ = 672,000 J ✓
- Mass evaporated, $m = Q/l = (12000)/(2.3 \times 10^6)$ ✓ = 0.0052kg = 5.2g ✓

Qualitative (Concept Test)

- What are the two possible consequences of heating or cooling a sample of material?
- If different masses of the same material are supplied with equal amounts of heat energy, how will the temperature increase depend on the mass?
 - Sketch a graph of temperature increase against mass.
- Define specific heat capacity as an equation and in words.
- What happens to the temperature of a substance as it changes from a liquid to a solid?
- Define, in words, the specific latent heat of vaporisation.
- What is the difference between latent heat of vaporisation and latent heat of fusion?

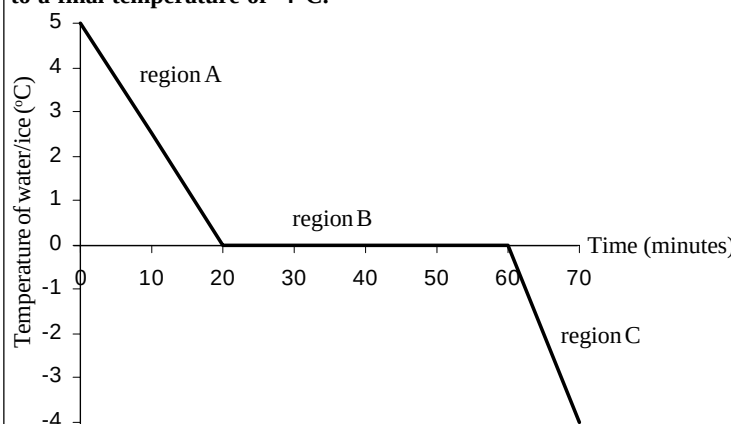
Quantitative (Calculation Test)

- What is the specific heat capacity of aluminium if it takes 18,200 J of energy to increase the temperature of a 5.0 kg block of aluminium by 4.0 °C? (2)
- How much heat energy will be supplied to 1.5kg of water when it is heated from 20°C to 80°C? (2)
(Specific heat capacity of water = 4,200 Jkg⁻¹°C⁻¹)
- How much heat energy will be given out by a block of copper with a mass of 2.0 kg, which is cooling down from 150 °C to 20 °C? Specific heat of copper = 390 Jkg⁻¹°C⁻¹ (2)
- How much heat energy will have to be supplied to melt 0.50kg of ice? Specific latent heat of fusion of ice = 330,000 J/kg. (2)
- What is the total amount of energy needed to heat up a copper kettle of mass 1.5 kg, from 20°C to boiling. The kettle is filled with water of mass 2.5 kg. (5)
 - Give 2 assumptions made in your calculation. (2)

Exam Workshop

This is a typical answer that might be given by a weak student in an examination. The comments explain what is wrong with the answers and how they can be improved. The examiner's answer is given below.

The graph below shows how the temperature is changing over a period of time for 25g of water that is being cooled, turned into ice, and taken to a final temperature of -4°C .



For this question you will need to know:

specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$;

specific latent heat of fusion of water/ice = $3.3 \times 10^5 \text{ J kg}^{-1}$.

- (a) In which region of the graph, A, B, or C, is the water changing into ice? (1)

C ✗ 0/1

In region C the water has completely turned to ice and has started to cool further. The temperature of all materials remains constant as they change state. The student should be looking for the region of the graph where the temperature does not change.

- (b) How much heat is extracted from the water to cool it during the first 20 minutes of the graph? (2)

$Q = mc\Delta\theta = (25)(4200)(5) \text{ ✗} = 525,000 \text{ J ✗}$ 0/2

The student has used the correct equation for specific heat capacity but has not converted the value for mass into kilograms and the value for time must also be converted into seconds.

- (c) How much heat energy is extracted from the water during the time that the temperature remains constant at 0°C ? (2)

$Q = ml = (25)(3.3 \times 10^5) \text{ ✓} = 8,250,000 \text{ J ✓ ecf}$ 2/2

The student has correctly used the equation but with the same mistake in the units of mass. This mistake would probably not be penalised a second time in the same question.

- (d) If a total of 210J of energy are extracted from the ice in the final 10 minutes shown by the graph, calculate the specific heat capacity of ice. (2)

$c = Q/m\Delta\theta = (210)/(25)(-4) = -0.21 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1} \text{ ✗}$ 0/2

The student has not given the heat removed a negative sign (as energy has been extracted from the ice). This has given an incorrect negative answer, which is also a thousand times too small because of the recurring mistake of quoting mass in g instead of kg.

Examiner's Answers

- (a) C
 (b) $Q = mc\Delta\theta = (0.025)(4200)(5) = 525 \text{ J}$
 (c) $Q = ml = (0.025)(3.3 \times 10^5) = 8,250 \text{ J}$
 (d) $c = Q/m\Delta\theta = (-210)/(0.025)(-4) = 2,100 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$

Quantitative Test Answers

- (1) $c = Q/m\Delta\theta = (18,200)/(5)(4) = 910 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$.
 (2) $Q = mc\Delta\theta = (1.5)(4200)(80-20) = 378,000 \text{ J}$
 (3) $Q = mc\Delta\theta = (2)(390)(20-150) = -101,400 \text{ J}$. Note the answer is negative, showing the copper block to be cooling.
 (4) $Q = ml = (0.5)(-330,000) = -165,000 \text{ J}$. Note the answer is negative showing that the ice is giving out energy and not being supplied with it.
 (5) (a) Energy needed to heat up water = $mc\Delta\theta = (1.5)(4200)(100-20) = 504,000 \text{ J}$
 Energy needed to heat up kettle = $mc\Delta\theta = (2.5)(390)(100-20) = 78,000 \text{ J}$
 Total energy needed = $504,000 + 78,000 = 582,000 \text{ J}$
 (b) The copper heats up to the same temperature as the water. There is no energy loss to the surroundings of the kettle such as air and the surface that it stands on.

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